INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
End Spring Semester Examination 2023-24

Date of Examination: 24.04.2024 Session: AN Duration: 3 hrs Full Marks: 60
Subject No: <u>AI61002</u> Subject: DEEP LEARNING FOUNDATIONS AND APPLICATIONS
Department/Center/School: Centre of Excellence in Artificial Intelligence
Specific charts, graph paper, log book etc., required: No
Special Instructions: Calculators are allowed. Rough work must be present in the answer script itself.

<u>Short Answer Questions</u> (Answer this as a *Separate* Section)

1. Which of the following propositions are true for a CONV layer? Justify. (Check all that apply.) [1]

(i) The number of weights depends on the depth of the input volume.
(ii) The number of biases is equal to the number of filters.
(iii) The total number of parameters depends on the stride.
(iv) The total number of parameters depends on the padding

2. Which of the following activation functions can lead to vanishing gradients? Give justification. [1]

(i) ReLU
(ii) Sigmoid
(iii) Leaky ReLU
(iv) None of the above

3. Give two benefits of using convolutional layers instead of fully connected ones for visual tasks. [1]

4. You are solving the binary classification task of classifying images as cat vs. non-cat. You design a CNN with a single output neuron. Let the output of this neuron be z. The final output of your network, y' is given by: [1]

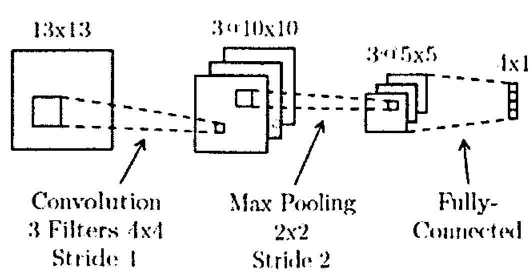
$$y' = \sigma(\mathrm{ReLU}(z))$$

You classify all inputs with a final value $y' \geq 0.5$ as cat images. What problem are you going to encounter?

5. In a 3-class CNN classifier, following the last FC-3 layer of your network, what activation must be applied? Given a vector a = [0.2, 0.2, 0.2], what is the result of using your activation on this vector? [1]

<u>Long Answer Questions</u> (Answer this as a *Separate* Section)

6. Below is a diagram of a small convolutional neural network that converts a 13x13 image into 4 output values. The network has the following layers/operations from input to output: convolution with 3 filters, max pooling, ReLU, and finally a fully-connected layer. For this network we will not be using any bias/offset parameters (b). Please answer the following questions about this network. (2.5 X 5)



A. How many weights in the convolutional layer do we need to learn?
B. How many ReLU operations are performed on the forward pass?
C. How many weights do we need to learn for the entire network?

D. True or false: A fully-connected neural network with the same size layers as the above network ($13\text{x}13 \rightarrow 3\text{x}10\text{x}10 \rightarrow 3\text{x}5\text{x}5 \rightarrow 4\text{x}1$) can represent any classifier that the above convolutional network can represent. Justify your answer.

E. What is the disadvantage of a fully-connected neural network compared to a convolutional neural network with the same size layers? Justify with numerical values.

7. With respect to CNN, answer the following: (2.5 X 5)

A. Design a 3x3 filter that detects (a) vertical edges in the image and (b) horizontal edges in the image

B. You are given a 3 X 3 input matrix M and 2x2 filter k below. Compute their dilated convolution with d = 2. Assume that gaps in the kernel are filled with zeros, stride is 1 and padding is 0. M = [1,2,3; 4,5,6; 7,8,9] and k = [1,1; 1,1] (Note: semicolon implies next row)

C. As compared to standard CNN layers for classification, which CNN layer helps in a regression task where the output dimension is similar to that of input. Explain how that layer works with an example. Also, mention which of the CNN layers for classification is detrimental for a regression task.

D. What is a Residual layer in a ResNet, and explain how it eliminates overfitting?

E. How do you perform activation maximization to visualize CNN performance?

(2.5 x 4)

8. a) What are the main challenges in training an RNN?

b) You are using a basic single layer unidirectional RNN based encoder-decoder architecture to translate sentences of a language with vocabulary size m to sentences in another language with vocabulary size n. The dimension of the hidden states in encoder and decoder are k and p respectively. How many parameters this model will have? Assume that there are no bias parameters.

c) In LSTM, under what condition the memory cell at t-th time point is a simple copy of the candidate memory cell of the same time point?

d) What is the role of layer normalization in transformer?

6+4

9. a) Consider a simple many-to-many RNN where the input sequence length equals to output sequence length. Let W be the parameter matrix associated with hidden state t-1 to hidden state t. Show that if the eigenvalues of W are greater than 1, then exploding gradient issue will appear. You must prove it mathematically. Assume *tanh* activation is used from hidden state t-1 to hidden state t.

b) Under what conditions an LSTM will behave like an RNN? Prove it.

5+5

10. a) You are using a transformer network for machine translation task. The transformer has m encoder and m decoder blocks with h attention heads in each block. Define how the encoder-decoder attention is computed. You must define it mathematically using keys, queries and values from encoder blocks and decoder blocks. No marks will be given unless you define it mathematically.

b) Design the BERT architecture for next sentence prediction task. Define each component of your architecture. No credit will be given if you draw only the diagram.